

TEAM AI MAVERICKS

USE CASE NO. 9 : FINANCE PLANNING

Meet Our Team

SANIKA KULKARNI

KASHISH JAIN

SARANG KOKANE

PRAJWAL KULKARNI

YATESH AHIRE

OMKAR KOKATE

UDAY LOWALEKAR

Problem Statement

Banking Finance Planning Using GenAI, customize finance plan for customers as per their profile. Profile will be built using current net worth, risk taking ability and financial goals. Using past trend data, couple of plans to be generated for user to choose from No reference required , Dummy data

Why Financial Planning Matters Today?

In today's dynamic world, many individuals struggle to align their spending, saving, and investing with their long-term aspirations. The path to financial well-being can seem daunting.

A personalized financial plan isn't just about numbers; it's about reducing stress, gaining clarity, and building the confidence needed to navigate life's financial journey successfully.



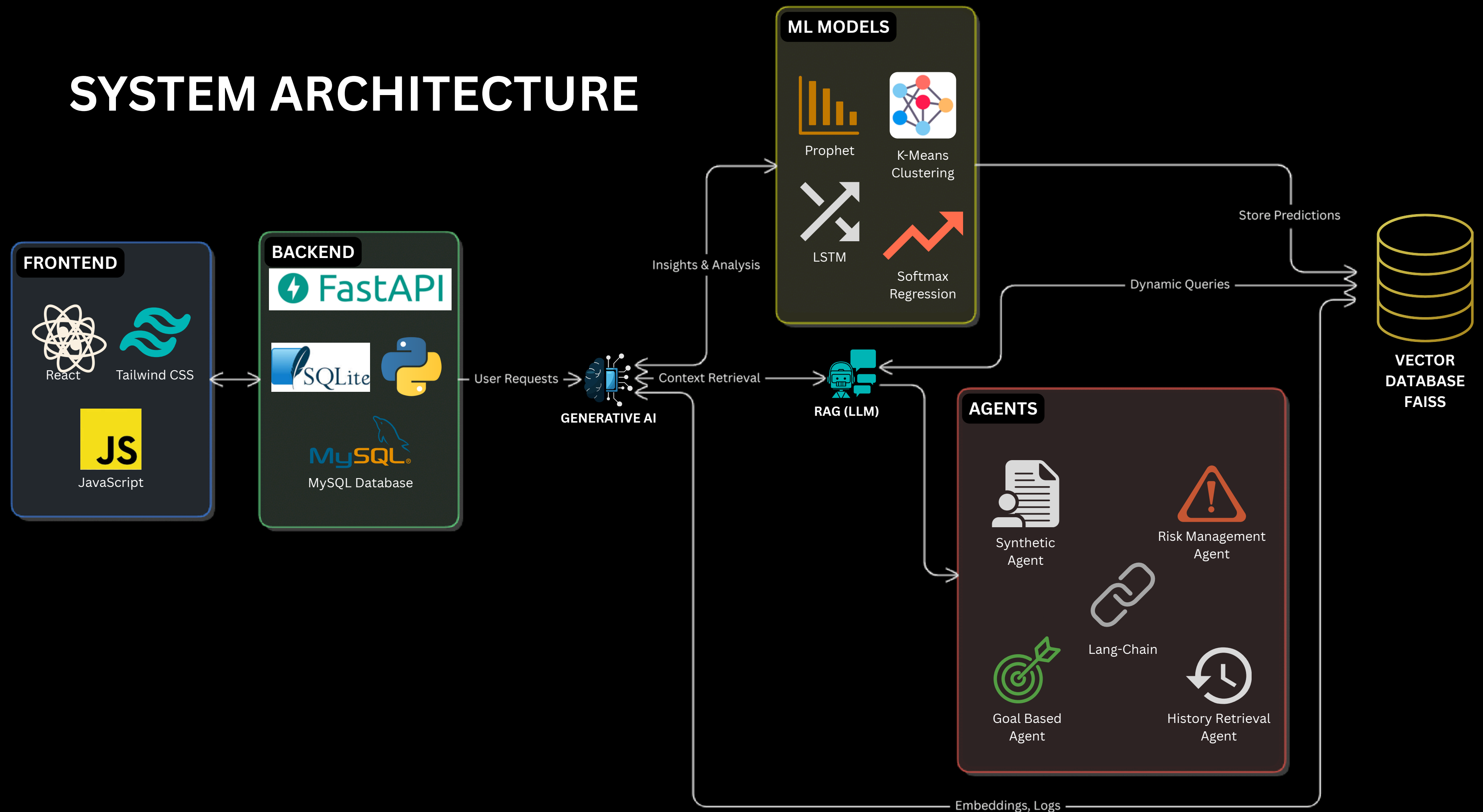
What is FinPlan?

FinPlan is a GenAI-powered financial planning app that creates personalized strategies based on a user's net worth, risk tolerance, and financial goals.

By analyzing past financial trends and market data, it generates multiple plan options—ranging from conservative to aggressive—so users can choose what suits them best. The app also provides clear AI-driven explanations, helping customers understand trade-offs and make confident financial decisions.



SYSTEM ARCHITECTURE



FRONTEND

React Application

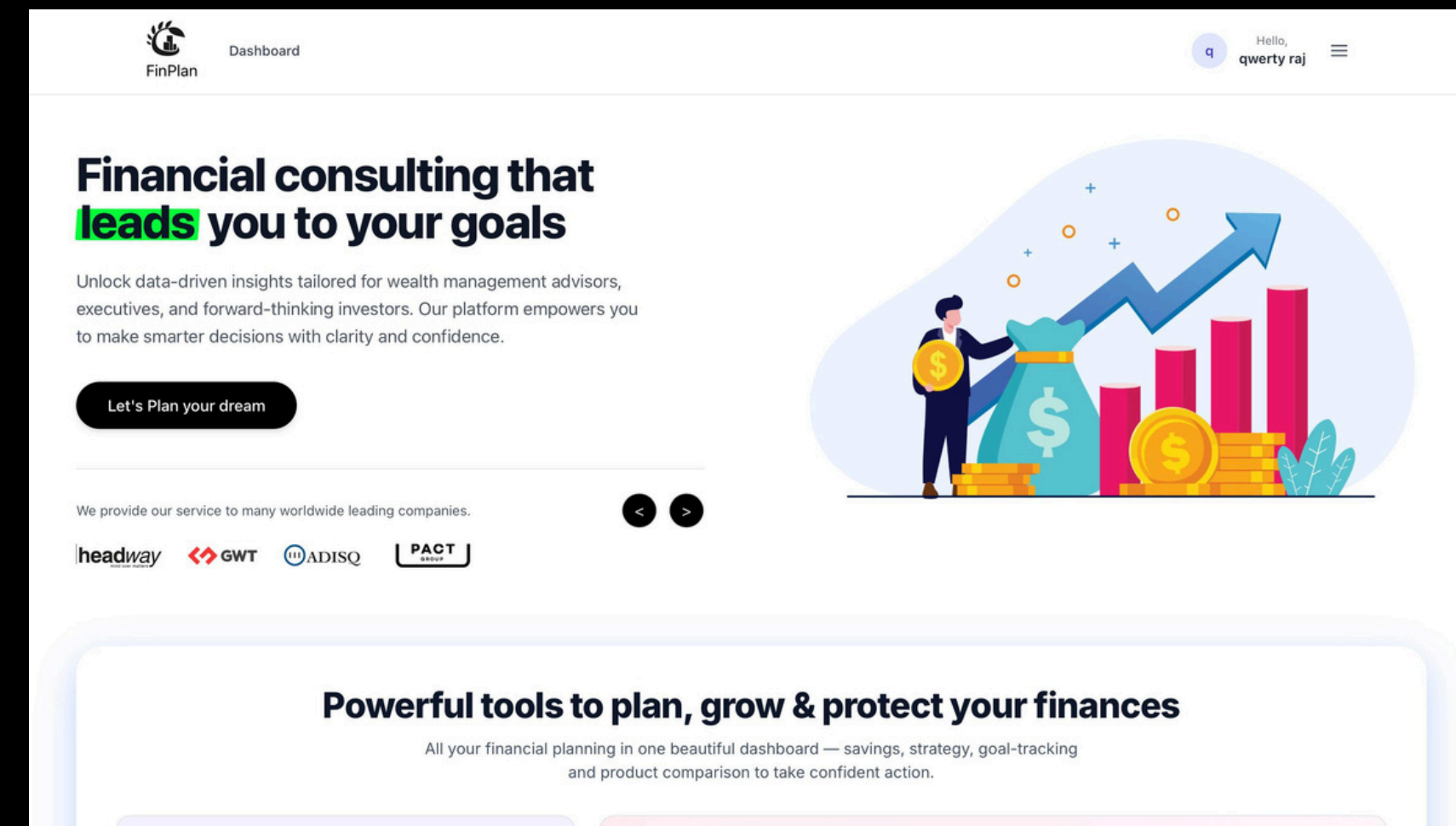
- Built with React, offering a modern, responsive UI.
- Connects to the backend via REST APIs.

Features:

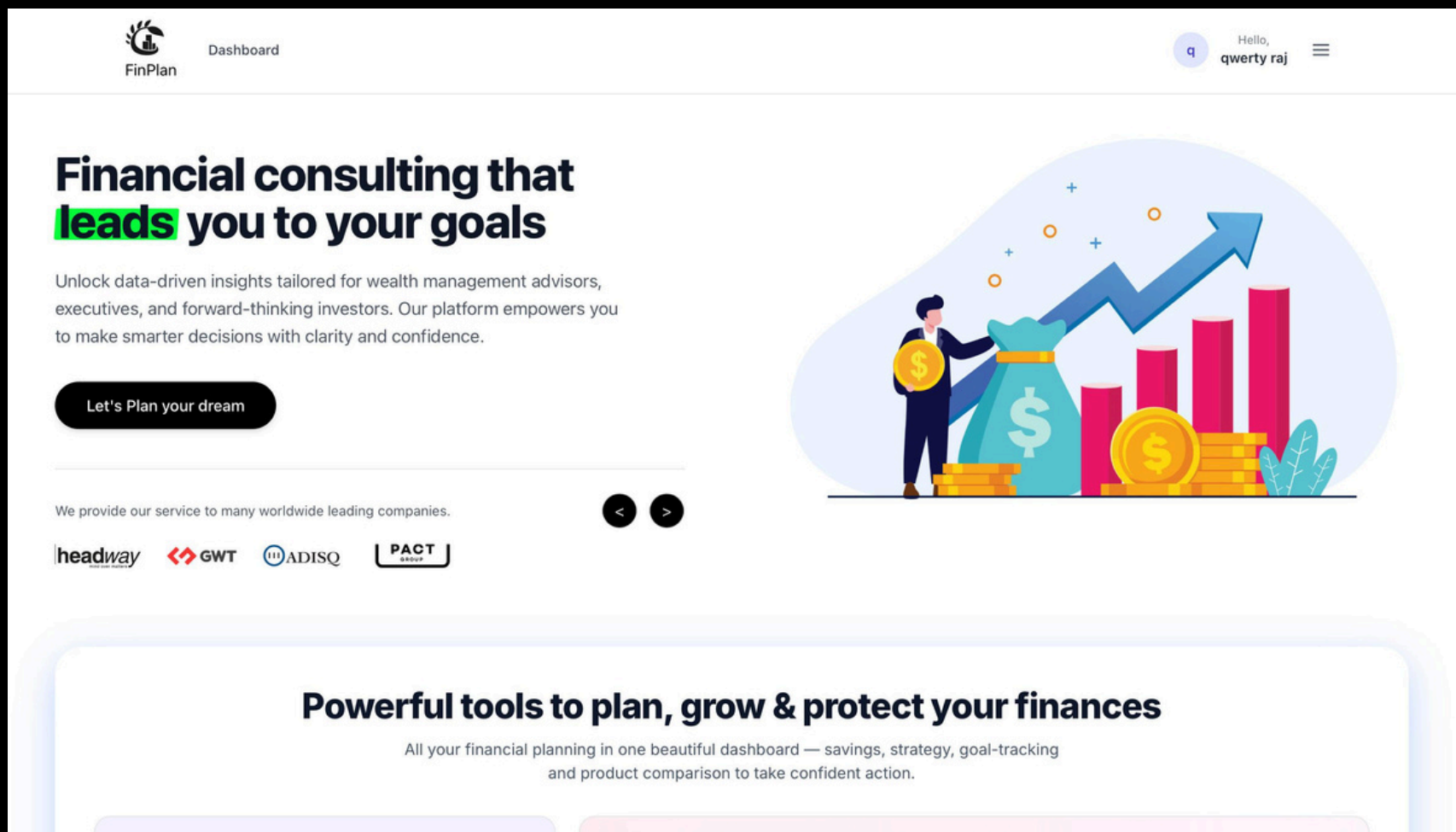
- User authentication with JWT tokens
- Interactive dashboards & charts
- Profile setup, investment tracking, and goal management
- AI insights integration with Streamlit

Tech Stack

React, Axios, Tailwind CSS, React Router, Lucide React

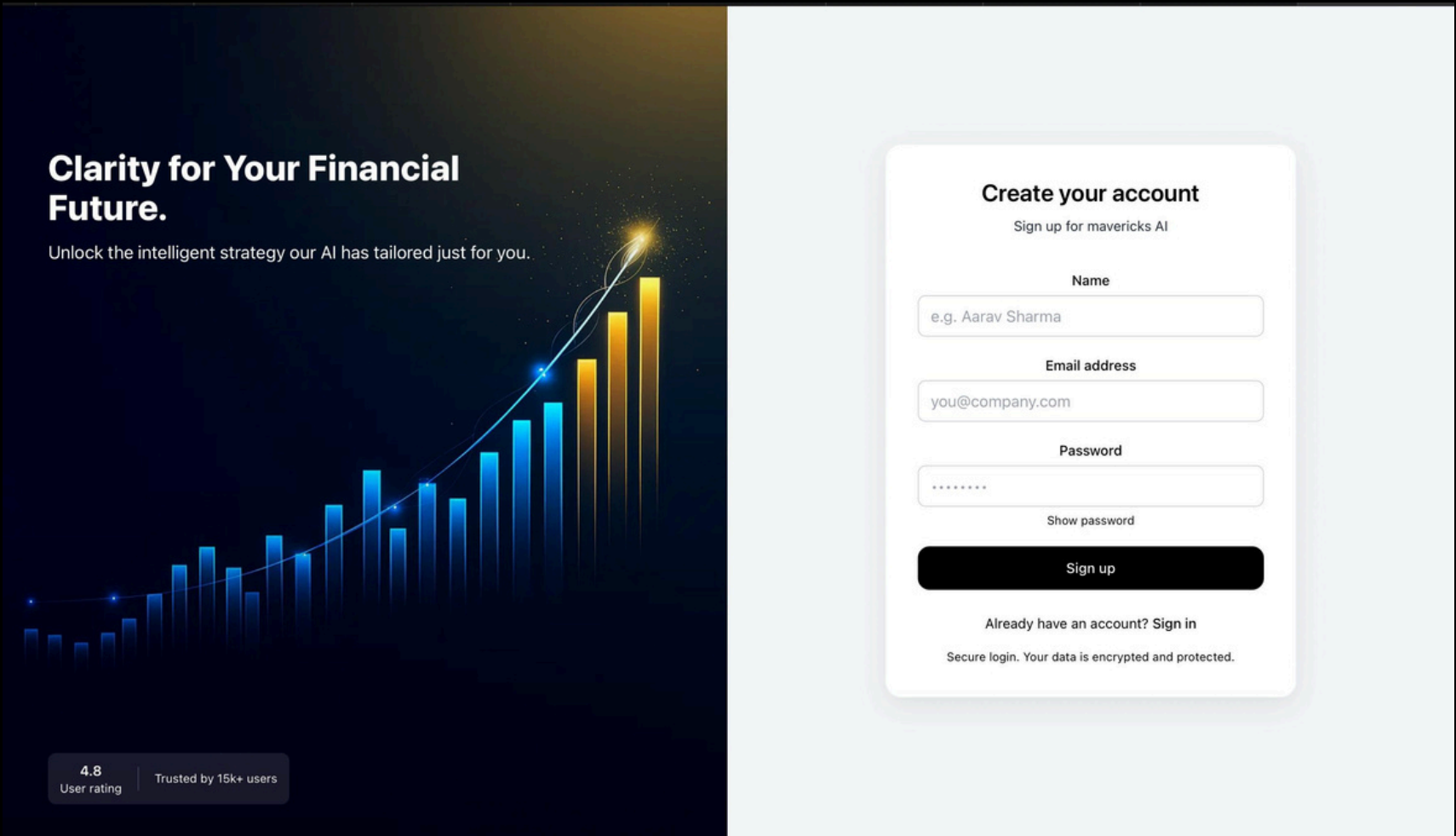


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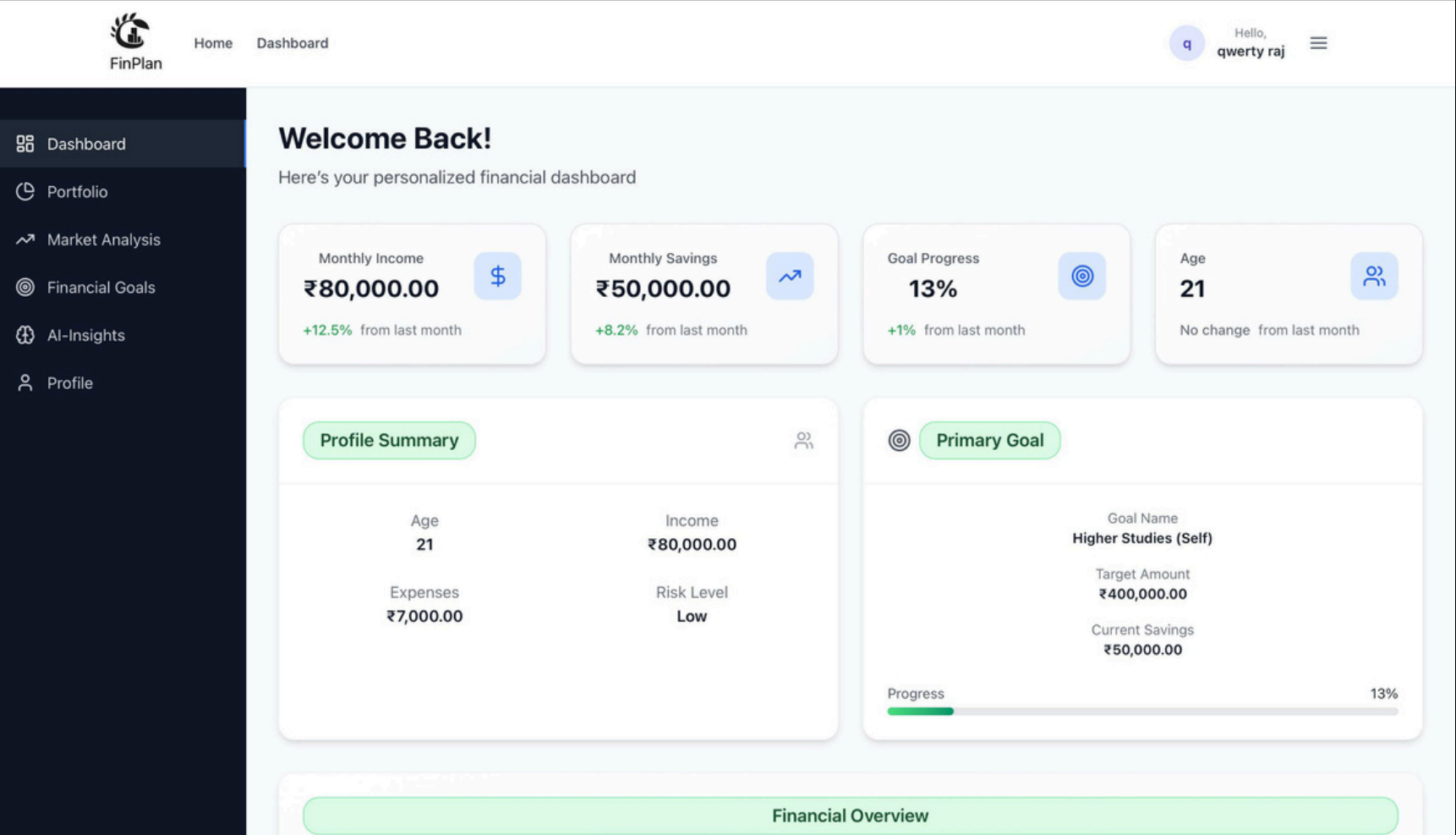


Login Page – Securely authenticate using email and password

Landing Page – Guides users to sign up or log in.

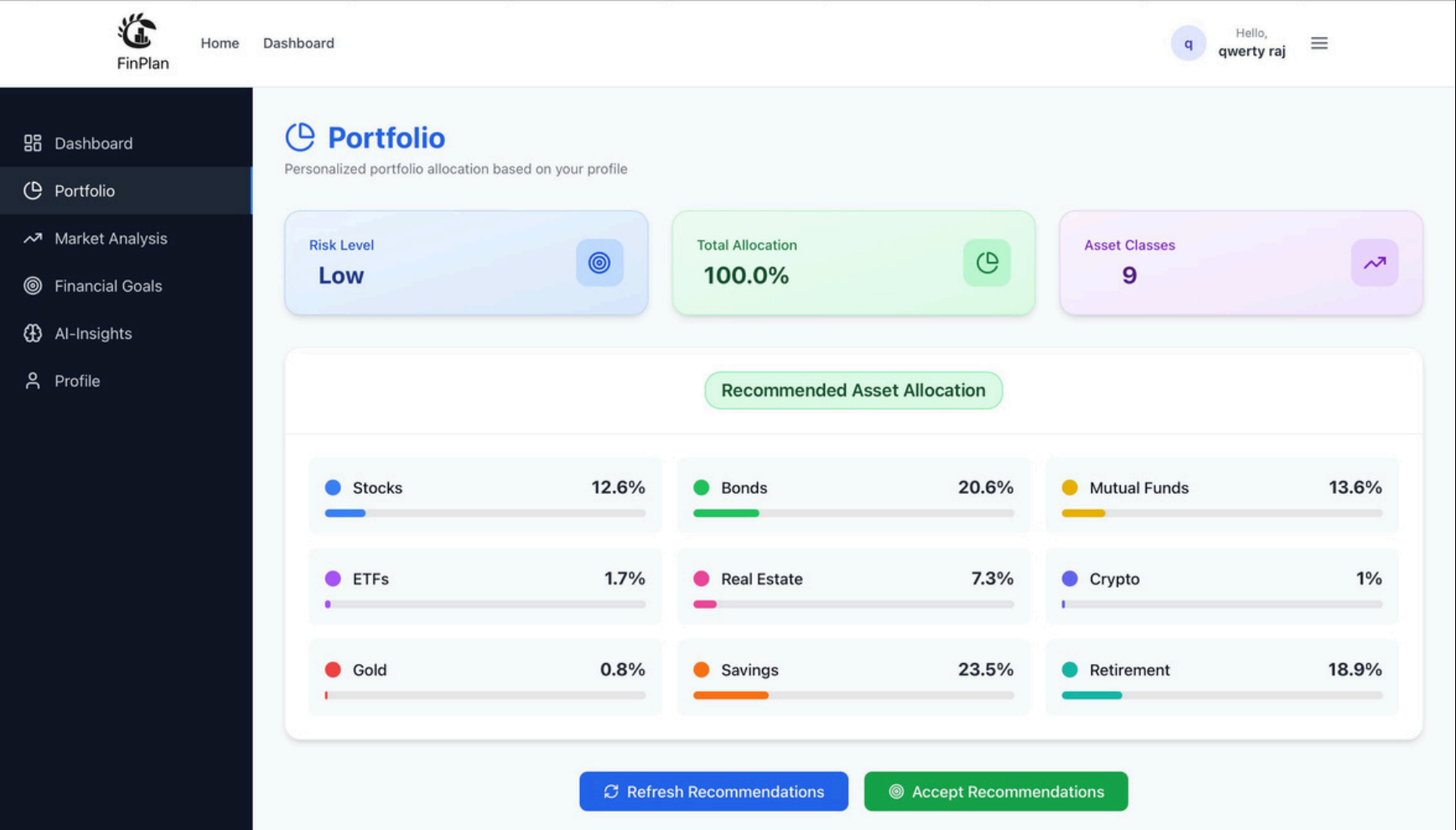


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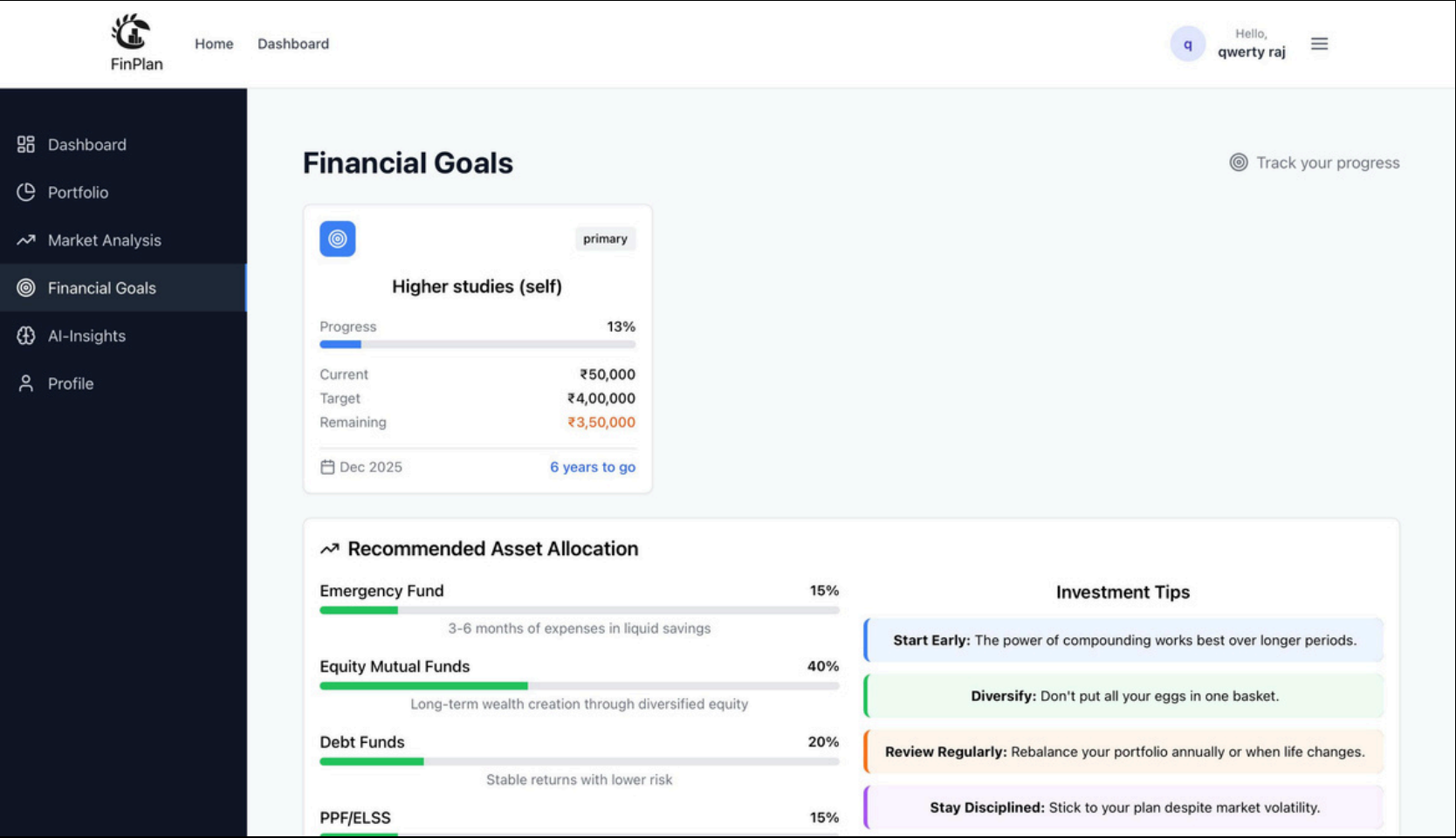


Portfolio Page – Helps users track and manage their recommended allocation

Dashboard Page – Displays key financial metrics and primary goal at a glance

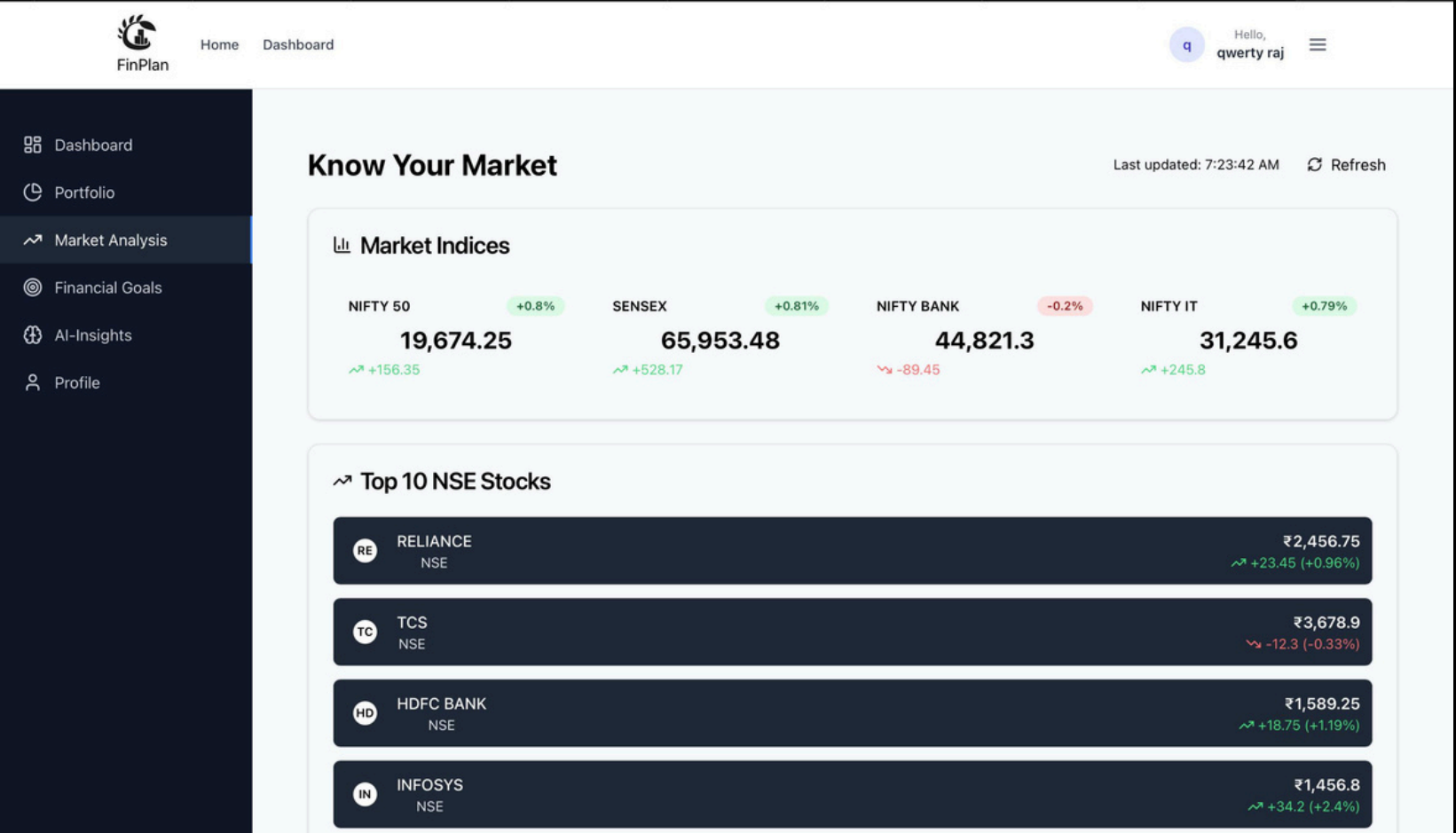


FRONTEND

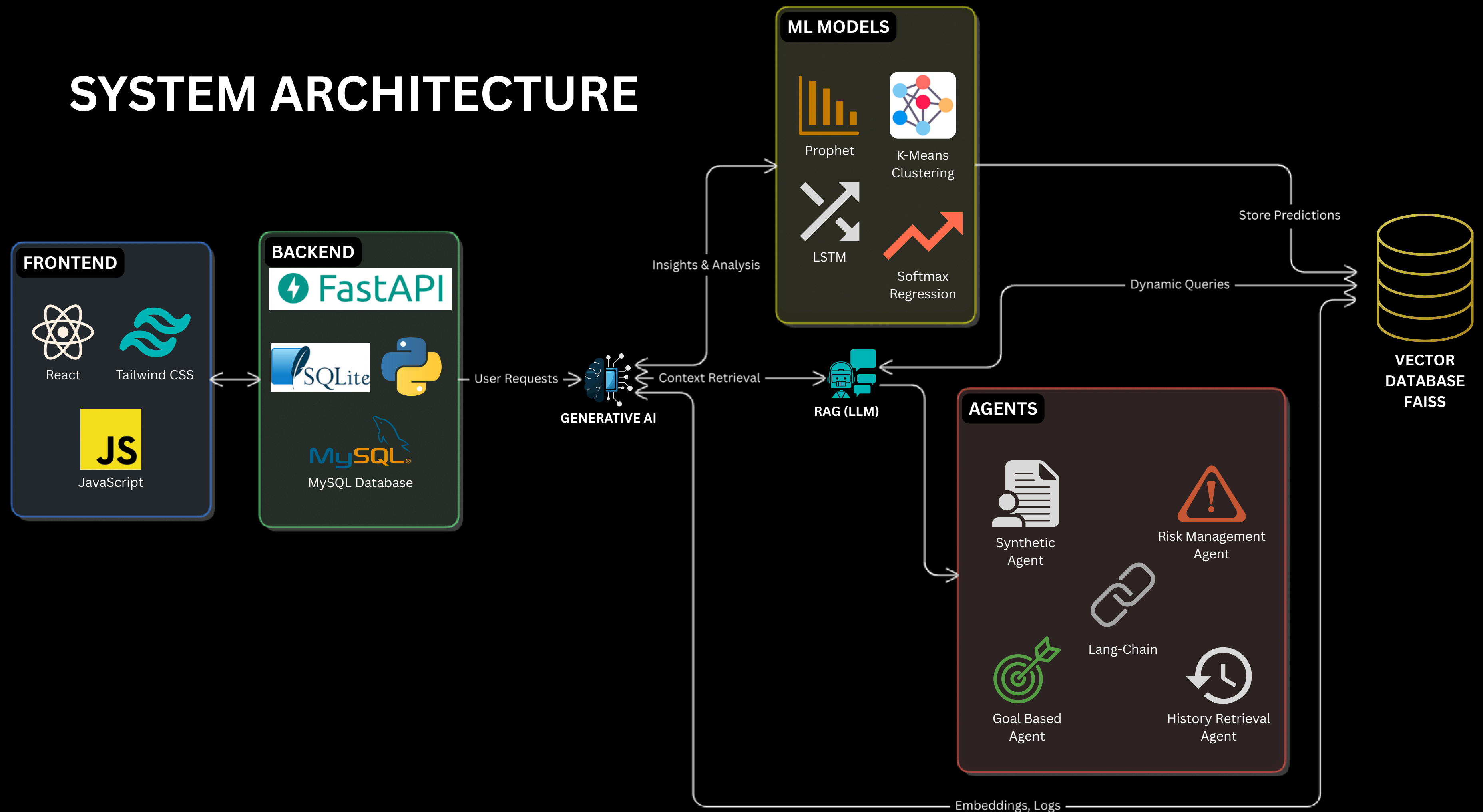


Market Analysis: Keeps the user updated with the latest information about the markets

Financial Goals Page: Lets user see their goal status



SYSTEM ARCHITECTURE



BACKEND

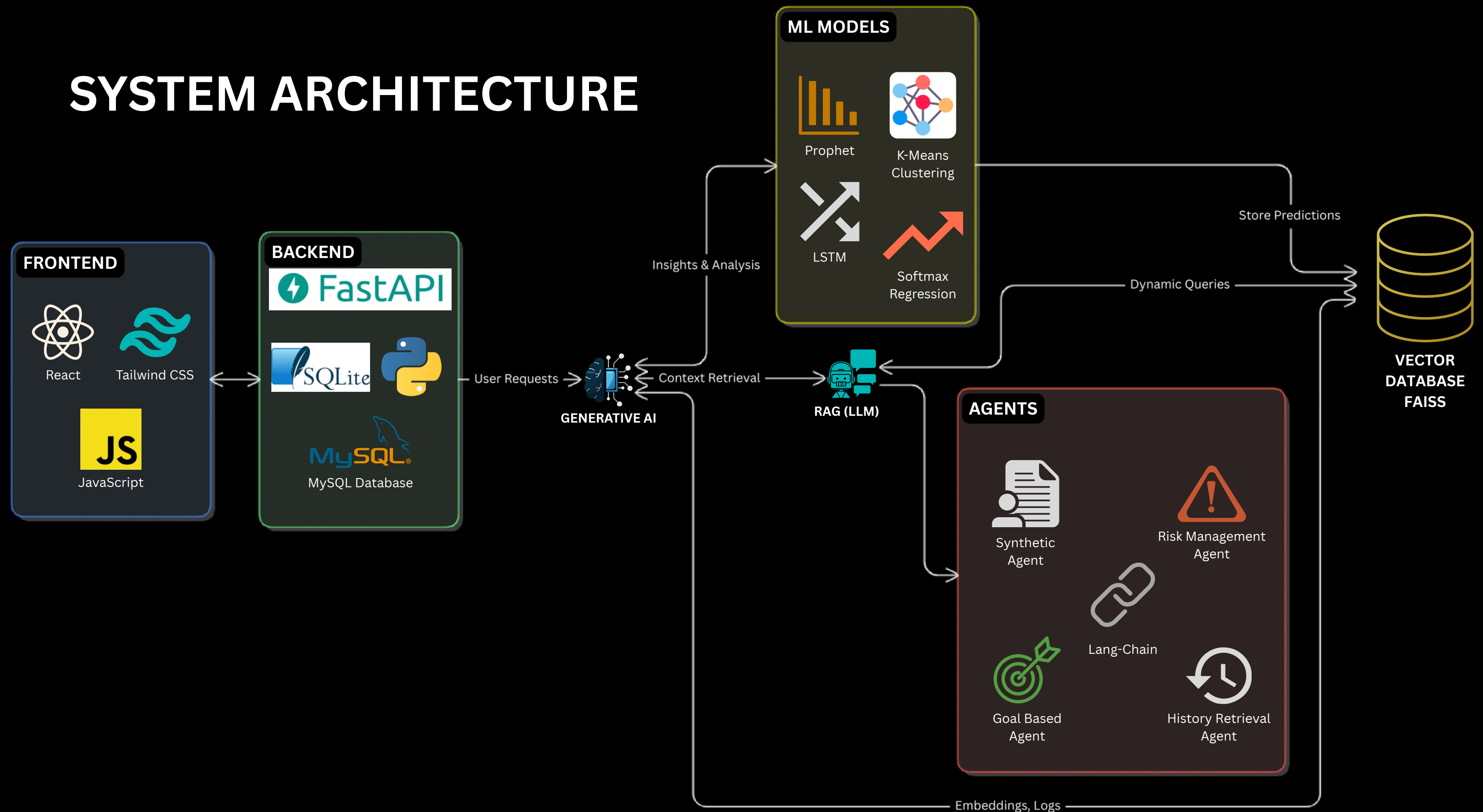
FinPlanAI Backend: Architecture & Stack

- **Backend:** FastAPI (Python) — single backend implementation.
- **Validation & typing:** Pydantic for request/response schemas.
- **ORM:** SQLAlchemy (clean DB models + migrations).
- API docs auto-generated at /docs.
- **Deployment target:** SQLite for dev, MySQL for production.
- **Speaker note (one-liner):** FastAPI provides a fast, typed, and maintainable backend with automatic docs and async support — the single source of truth for all business logic.

Core Services & Request Flow

- **Main routers:** auth, profile, dashboard, portfolio, me.
- **Auth endpoints:** POST /auth/signup, POST /auth/login, POST /auth/refresh, POST /auth/logout.
- Business logic separated into services/ (e.g., auth_service.py, profile_service.py) for testability.
- **Middleware:** JWT validation extracts user context for protected APIs.
- **Typical flow:** React frontend → FastAPI router → service layer → SQLAlchemy → DB → response.
- **Key tables:** user_accounts, user_profiles, refresh_tokens, financial_goals, transactions

SYSTEM ARCHITECTURE



DATA GENERATION

Why Data Generation?

- Sourced dataset had only 40 records – too small for ML.
- Needed a larger, realistic dataset for reliable modeling.

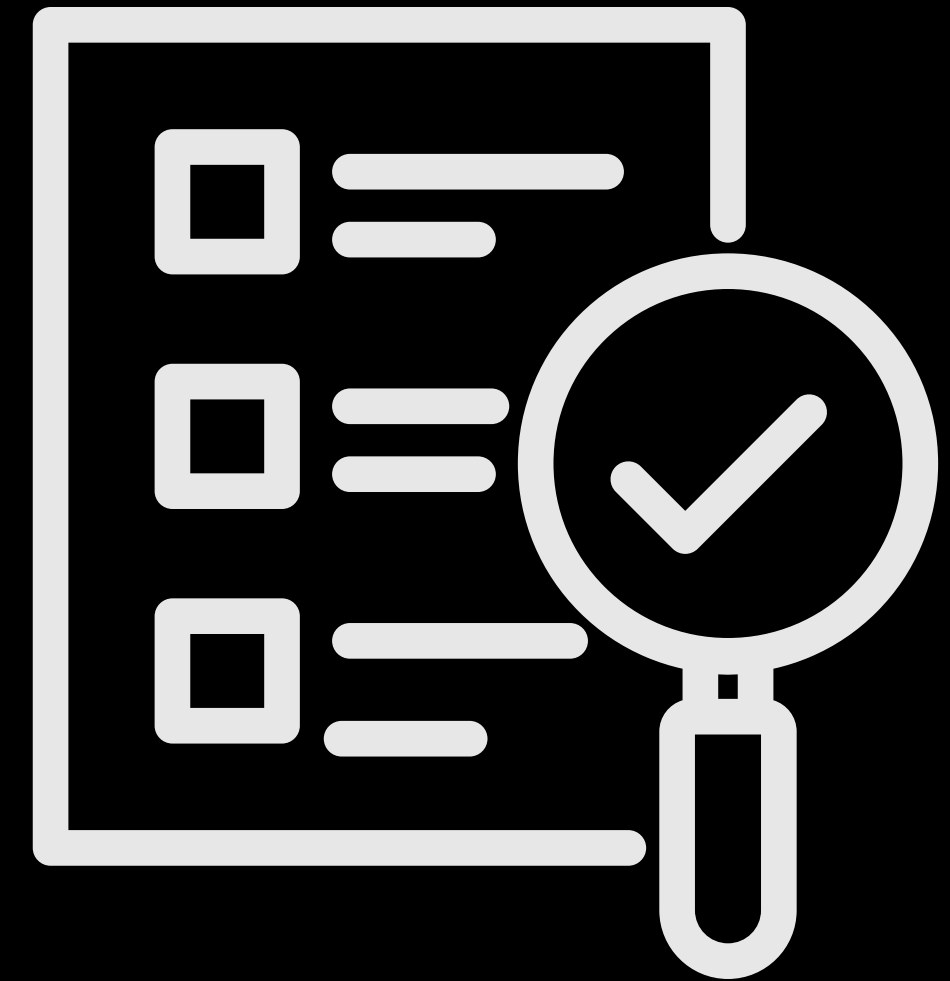
Approach:

1. Survey & Research References:

- RBI Household Financial Survey (savings patterns)
- SEBI Investor Survey (risk appetite by age)
- World Gold Council Reports (gold investment trends)
- NSE Investment Behavior Study (portfolio diversification)

2. Generation Process:

- Created about 6,000 records.
- Followed normal distribution for numerical features.
- Maintained real-world ratios from surveys.
- Added 5% noise for realistic variation.
- Preserved logical relationships (e.g., high income → higher savings).



ML MODELS

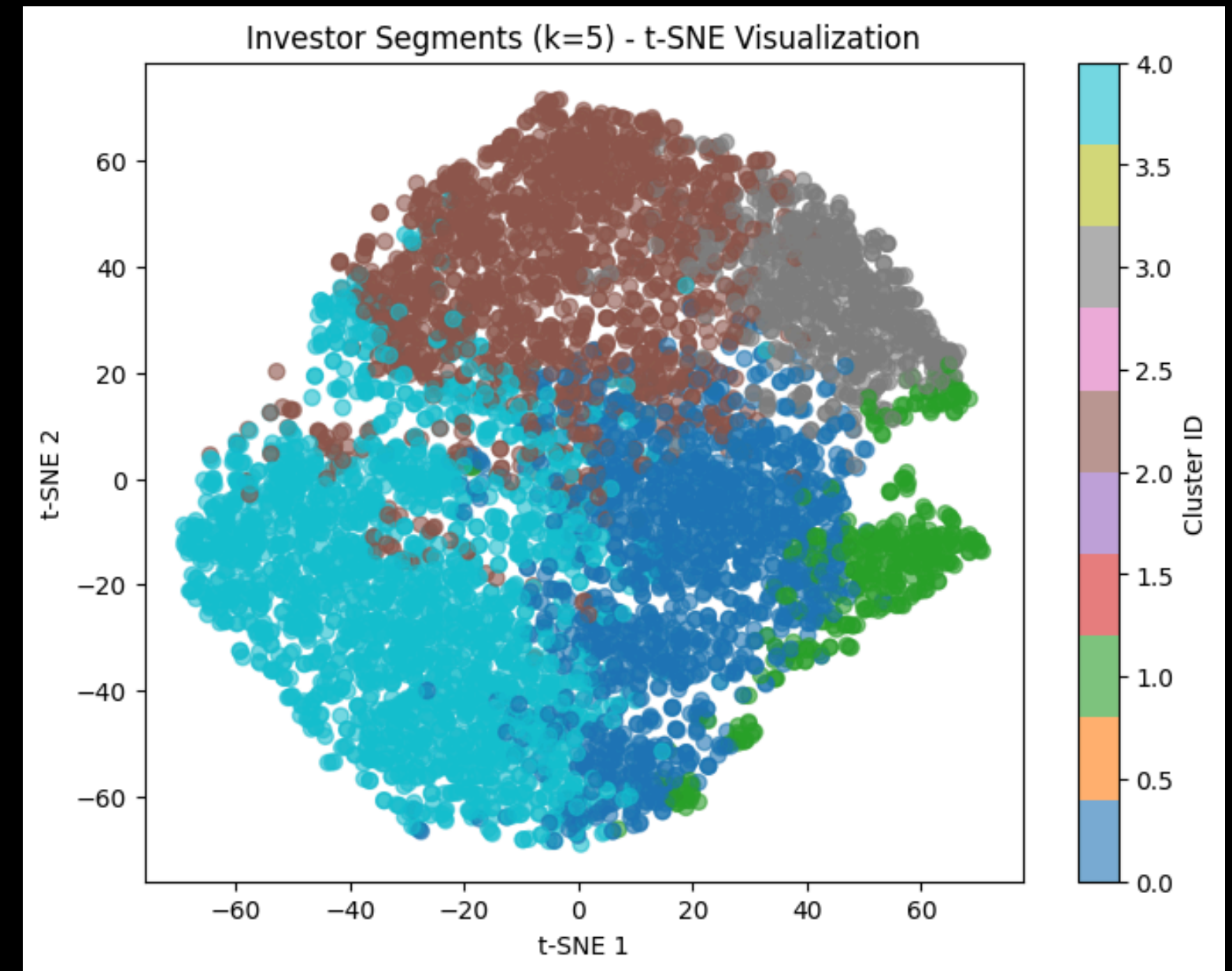
1. K-Means Clustering

Purpose:

- Group investors with similar financial profiles for targeted investment strategies.

Process:

- Feature Selection: Income, Savings, Debt, Risk Score, Financial Goal, etc.
- Preprocessing: Data normalization for scale consistency.
- Cluster Selection: Optimal k determined using Elbow Method.



Results:

Young Investors

Late-20s, low income/net worth, long horizon. Highest risk appetite, prefer stocks & crypto for growth.

Wealthy Professionals

Mid-40s, high income/net worth, balanced portfolios with more bonds. Moderate risk, stability-focused.

Moderate Wealth, Mid-age

Late-40s, moderate wealth. Conservative, favor bonds & retirement; capital preservation priority.

Working Class, Mid-age

Mid-30s, moderate income/debt. Growth-focused mix with stocks, ETFs, some crypto; accepts moderate risk.

Older Investors

Around 60, high net worth. Safe assets (bonds, gold, retirement funds), short horizon.

ML MODELS

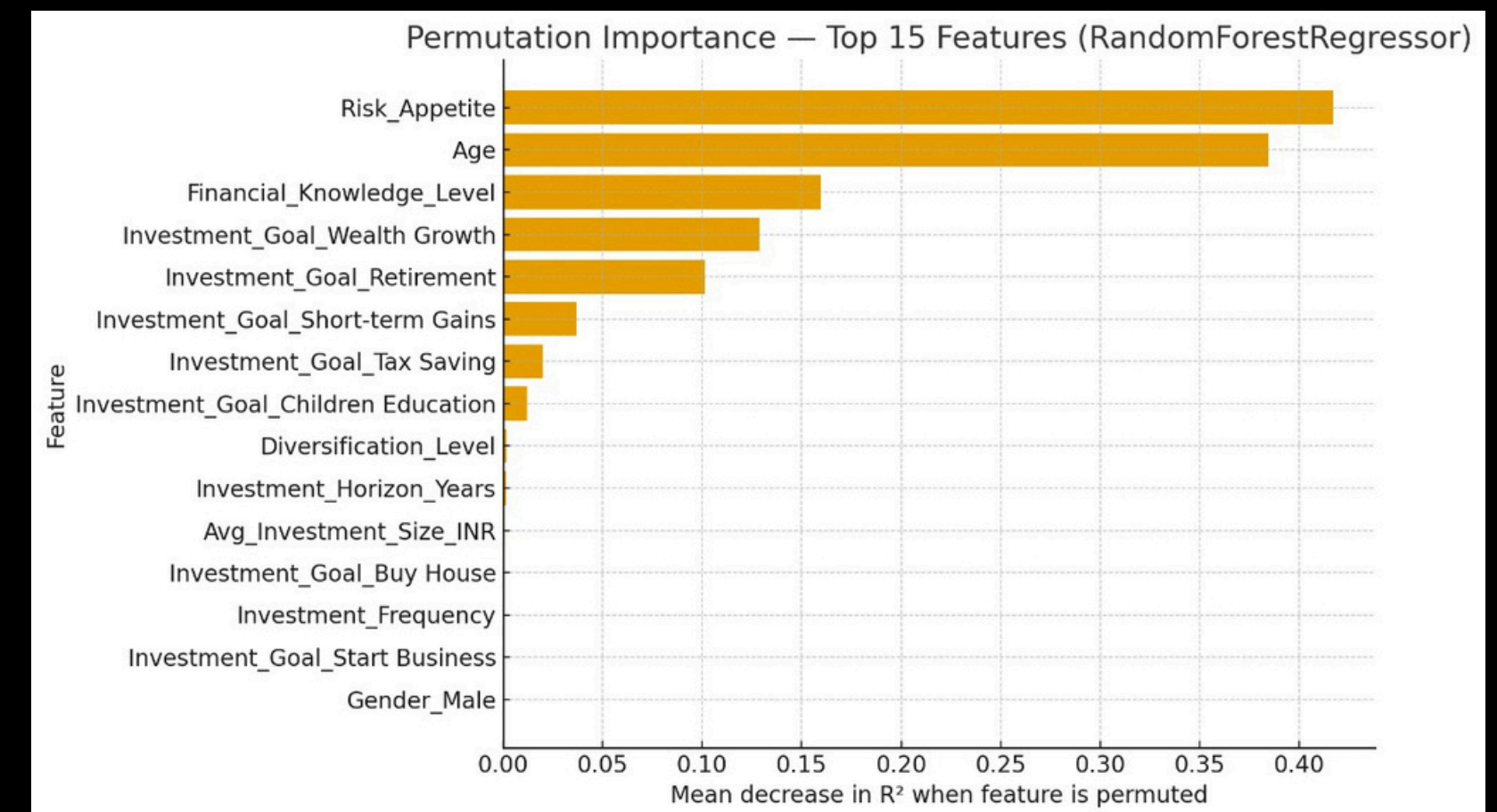
2. Softmax Regression

Purpose:

- ML model for personalized investment allocation based on user profiles and goals.

Process:

- Feature Selection: Income, Savings, Debt, Risk Score, Financial Goal, etc.
- Built a multi-output regression neural network with a Softmax activation layer
- Validated model performance on unseen data and tested with user profiles.

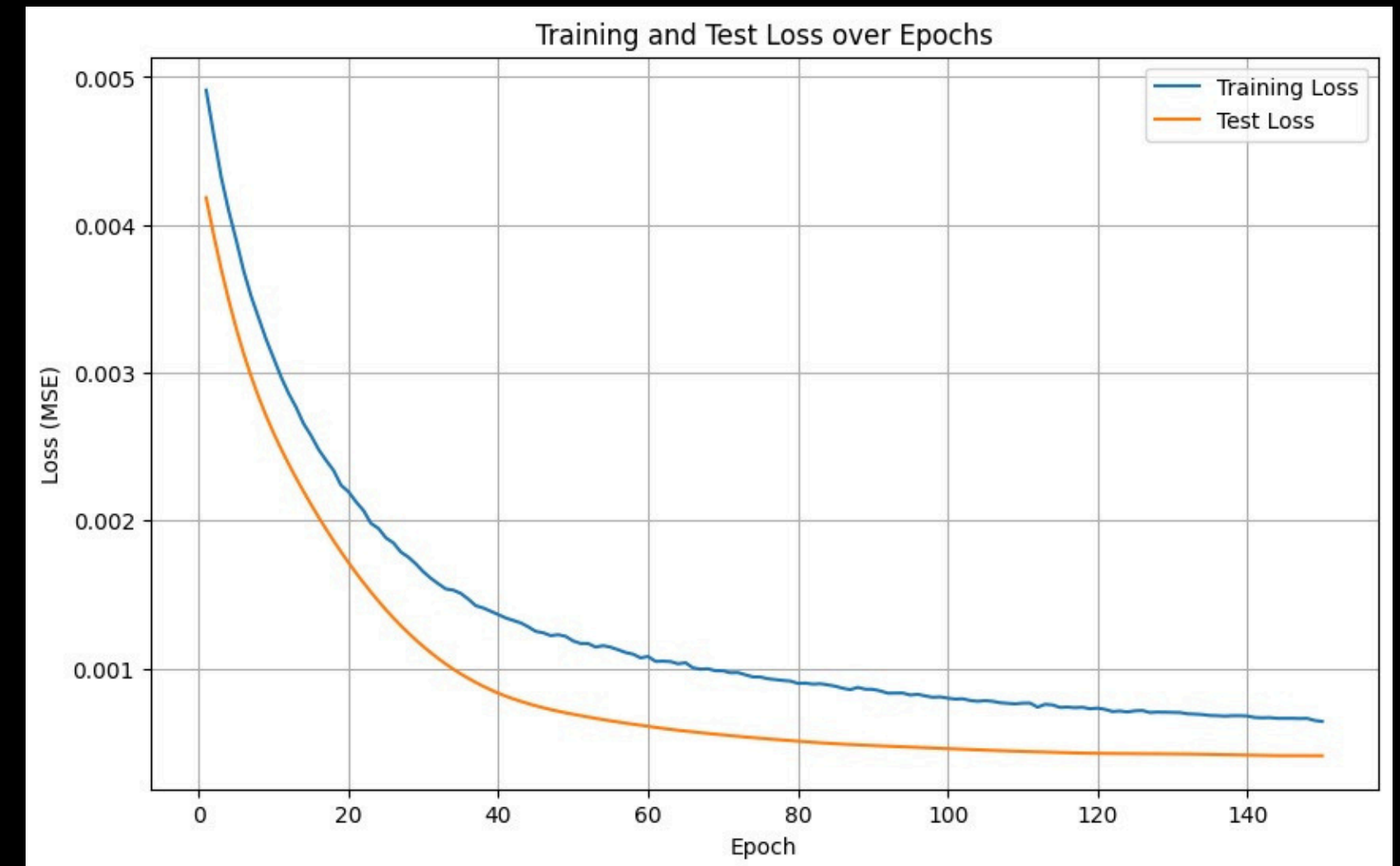


Results:

Model predicts valid, 100%-summing portfolio allocations.

Demonstrated strong generalization on test data.

Produced diversified portfolios consistent with different risk appetites and investment goals.



ML MODELS

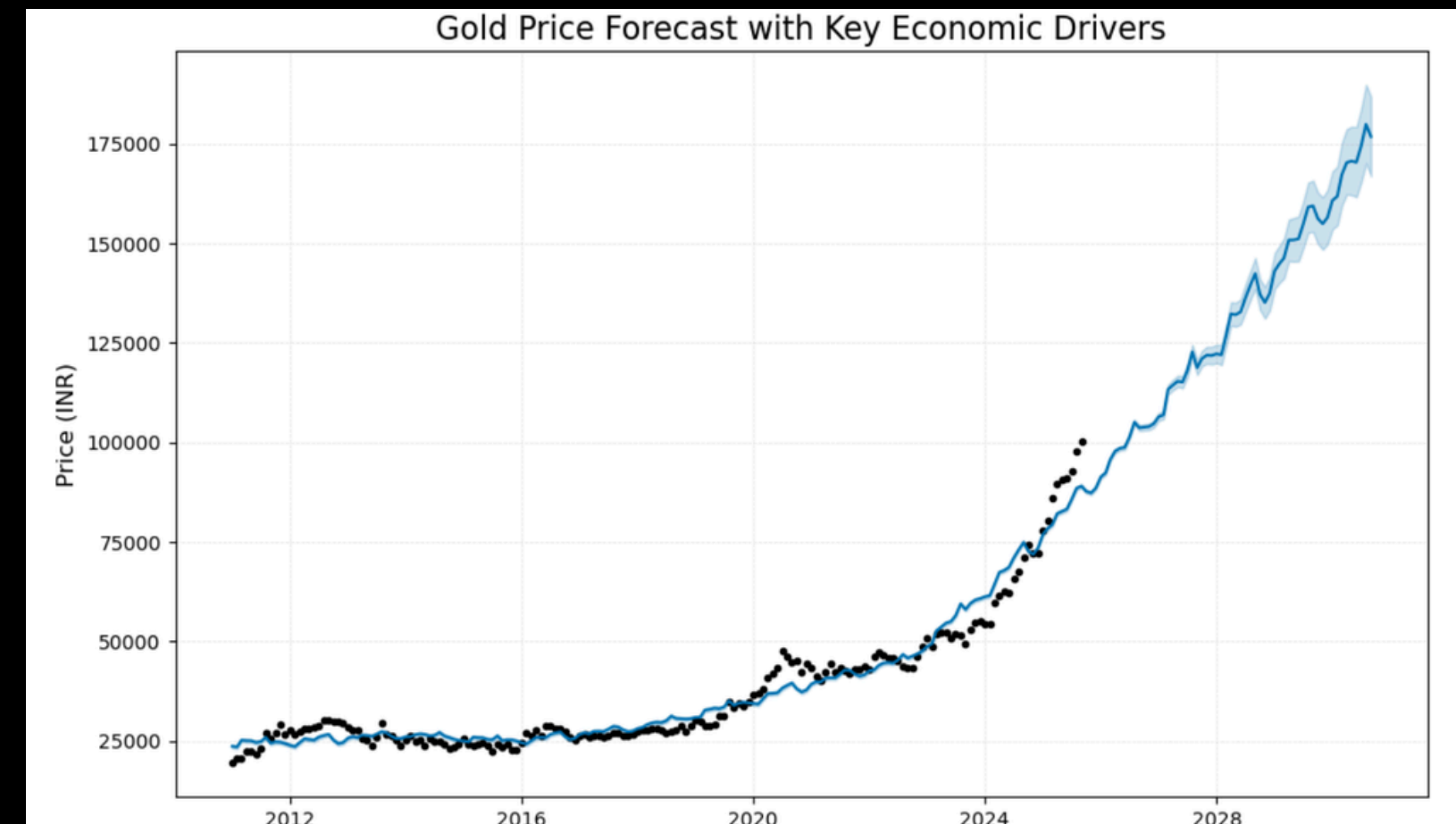
3. Prophet

Purpose:

- To forecast the price of gold in Indian Rupees (INR) for the next five years.

Process:

- Forecast the Drivers: The script begins by creating three separate 5-year forecasts for the key economic drivers: the US gold price, the USD/INR exchange rate, and Indian inflation.
- Train the Main Model: It then trains a primary model on the historical data.
- Generate Final Forecast: Finally, it feeds the future predictions of the drivers into the main model to produce the final, more accurate 5-year forecast.

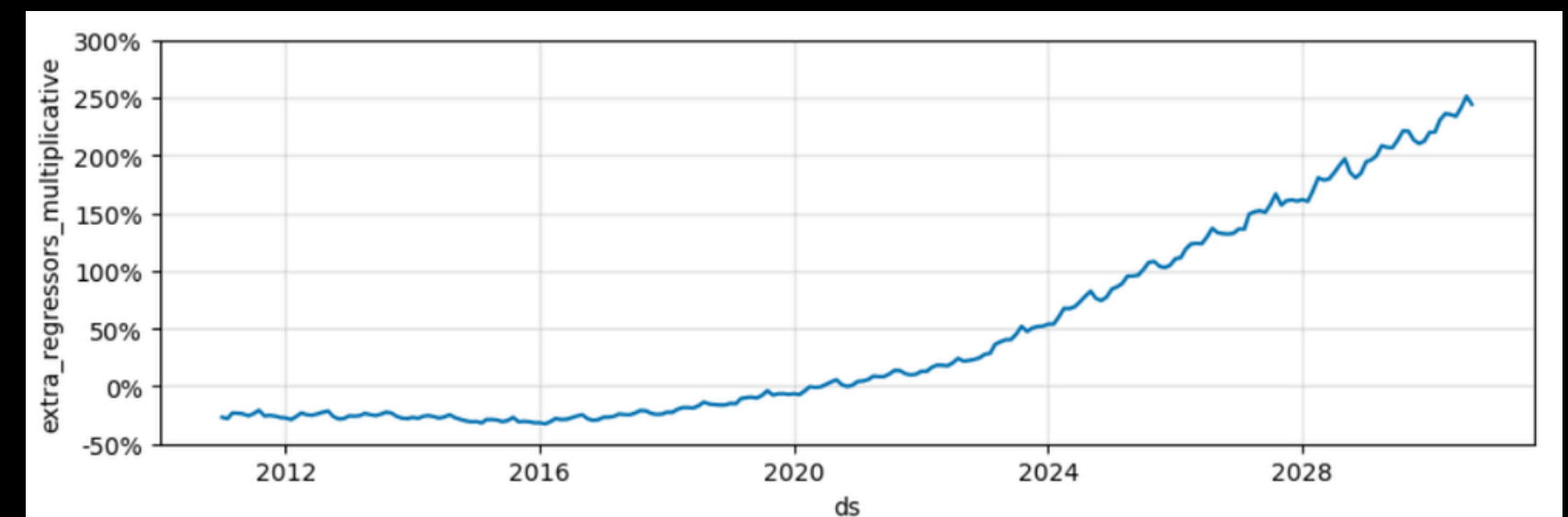
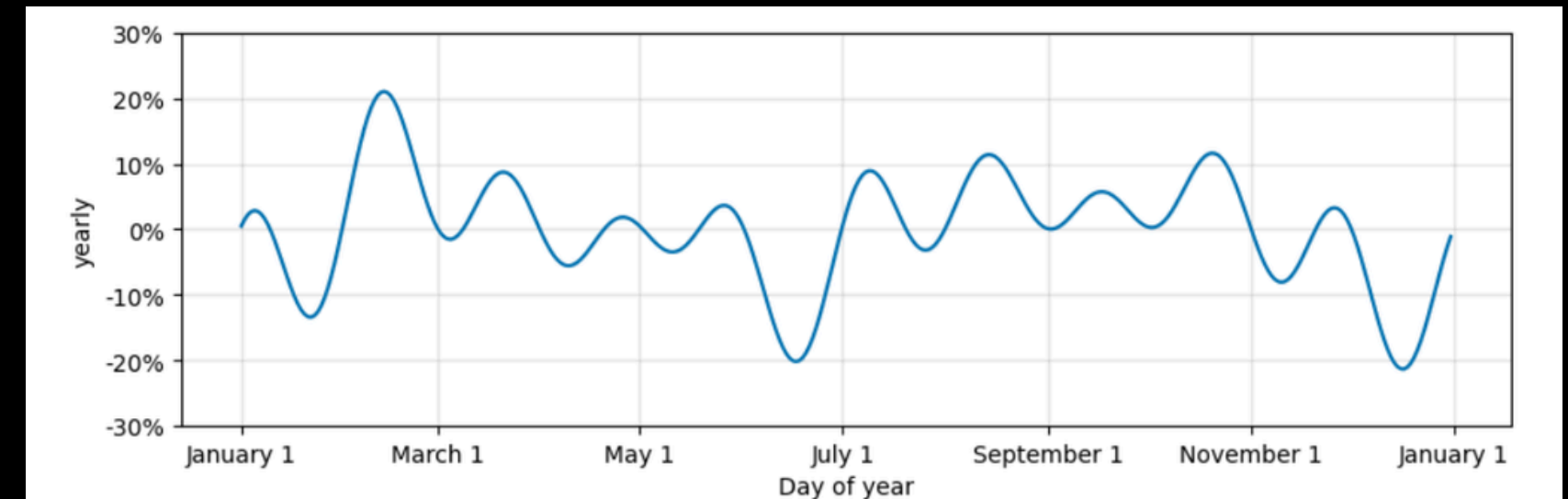
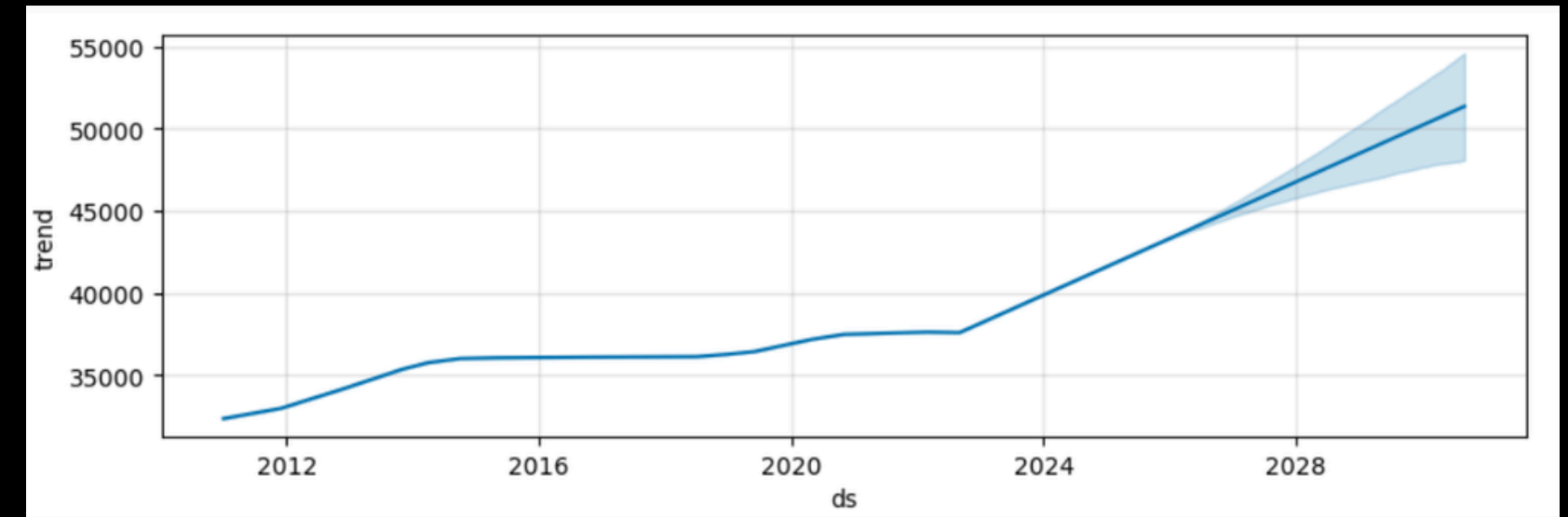


Results:

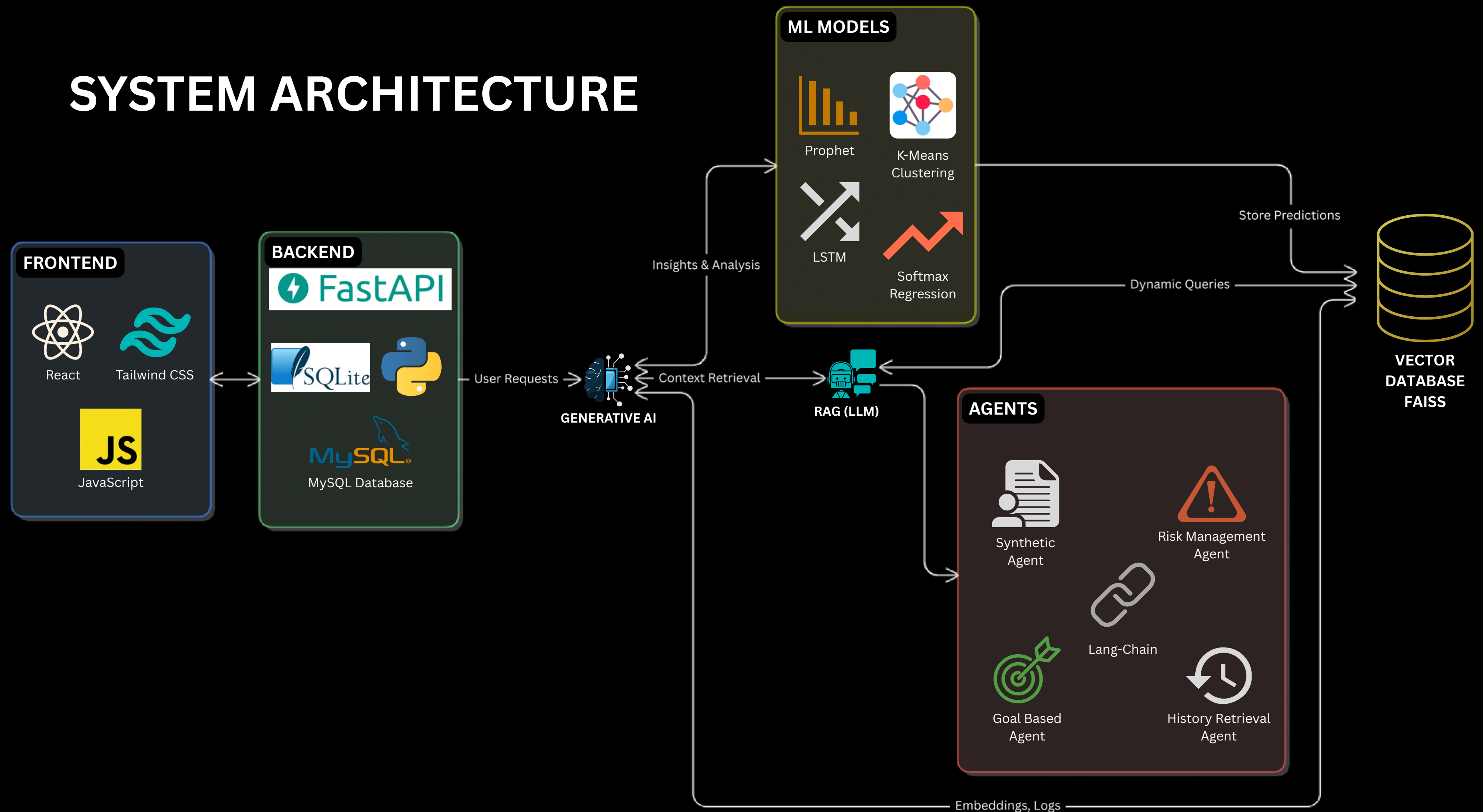
The model forecasts a strong and continuous increase in the price of gold.

The model successfully identified and quantified a yearly seasonal pattern in gold prices.

The model provides a clear "cone of uncertainty" (yhat_lower and yhat_upper). This quantifies the forecast's risk, showing that while the trend is upward, there is still significant potential for price fluctuation.



SYSTEM ARCHITECTURE



GENERATIVE AI

Usefulness in Problem Statement (PS):

- Traditional financial tools give generic, one-size-fits-all advice.
- GenAI enables personalized, conversational financial planning by understanding natural language queries and tailoring responses to each customer's profile.

Implementation in This Project:

- GenAI is integrated with RAG (Retrieval-Augmented Generation), Agents (Risk, Goal, History, Synthesis), and a Vector Database (FAISS).
- This allows the LLM to access customer KYC, investment documents, and financial data for generating grounded, domain-specific advice.

Gemini Models Used:

- Gemini 2.5 Pro – advanced reasoning, deep contextual analysis, plan synthesis.
- Gemini 2.5 Flash – lightweight, faster responses for conversational queries.



Advantages of Using GenAI in This Project:

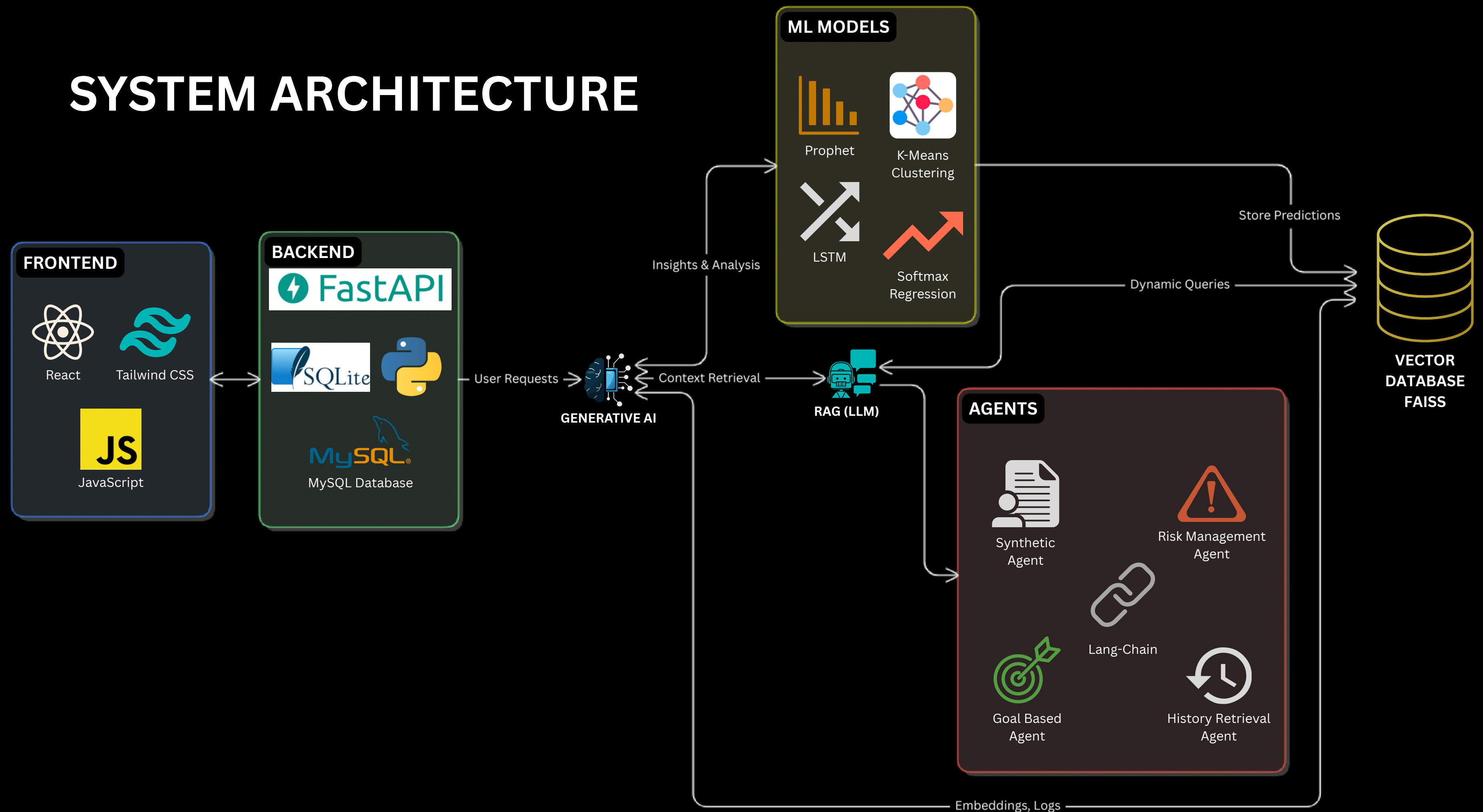
Contextual Awareness – Remembers history, understands customer goals.

Interactivity – Provides human-like, interactive planning instead of static reports.

Detailed Advice & Explanation – Breaks down recommendations (debt, SIP, insurance) step by step.

Adaptability – Works with varied financial profiles, scales to new goals and scenarios.

SYSTEM ARCHITECTURE



RAG (LLM)

What is RAG?

- RAG = Retrieve + Augment + Generate.
- It retrieves relevant knowledge from external documents, augments the user query with this context, and then lets the LLM generate accurate, grounded responses

Implementation in This Project:

- LangChain orchestrates the pipeline: embedding creation, retriever, and conversational chain.
- FAISS Vector Database is used to store embeddings of financial documents for fast and scalable retrieval.

Documents Used in Retrieval:

- Customer KYC forms
- Loan PDFs & Agreements
- Policy documents (insurance, investment rules)
- Market trend reports
- These ensure that every generated plan is based on real, domain-specific evidence.



Advantages of RAG in Banking & Finance:

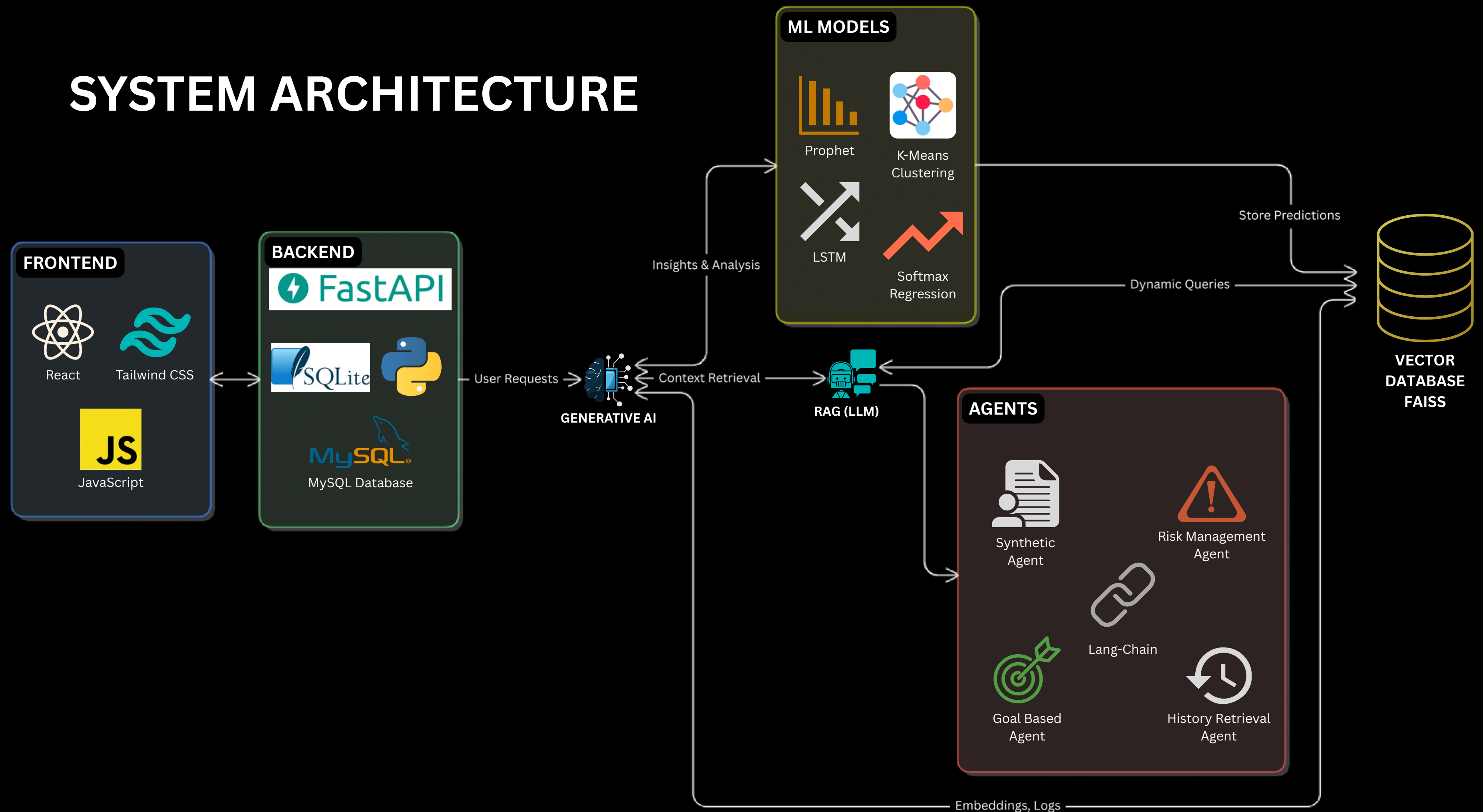
Avoids hallucinations by grounding LLM responses in verified documents.

Optimized for banks/financial firms → scalable, efficient retrieval from large document sets.

Context-aware responses tailored to each customer's financial profile.

Regulatory compliance → ensures plans consider real policies & updated guidelines.

SYSTEM ARCHITECTURE



AGENTS

Why Agents over Static Chatbots?

- Traditional desk chatbots are scripted and limited.
- Smart agents provide dynamic decision-making, acting like a “brain” that can answer queries, fetch documents, run financial math, or generate natural language explanations.

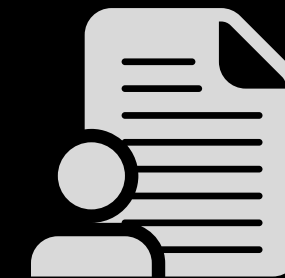
Beyond Static RAG:

RAG retrieves documents, but agents decide when and how to act. They can:

- Perform calculations,
- Recall conversation history,
- Explain financial policies step-by-step.

LangGraph Orchestration:

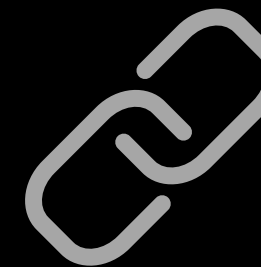
- LangGraph is used to coordinate multiple agents in a workflow.
- It ensures agents can communicate, share memory, and execute tasks in sequence or parallel, giving a modular and scalable architecture.



Synthetic
Agent



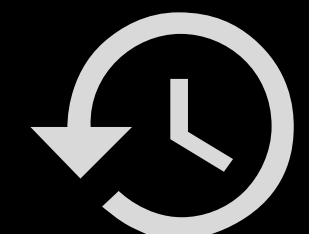
Risk Management
Agent



Lang-Chain



Goal Based
Agent



History Retrieval
Agent

Agents Implemented in This Project:

History Agent – remembers user's past chats.

Goal Agent – extracts financial goals.

Risk Agent – categorizes user risk profile (Conservative/Balanced/Aggressive).

Synthesis Agent – merges all insights into a coherent financial plan.

HOW EACH AGENT WORKS?

1. Goal Agent

Purpose: Understands the user's objective or intent (e.g., financial planning, insurance inquiry).

Role: Interprets queries, sets the direction for other agents, and ensures the system focuses on the user's desired outcome.

2. Risk Agent

Purpose: Assesses risks related to the user's query (e.g., loan eligibility, insurance risk).

Role: Analyzes data for risk factors, provides risk-related insights, and flags potential issues.

3. Synthesis Agent

Purpose: Combines information from multiple sources or agents to generate a comprehensive answer.

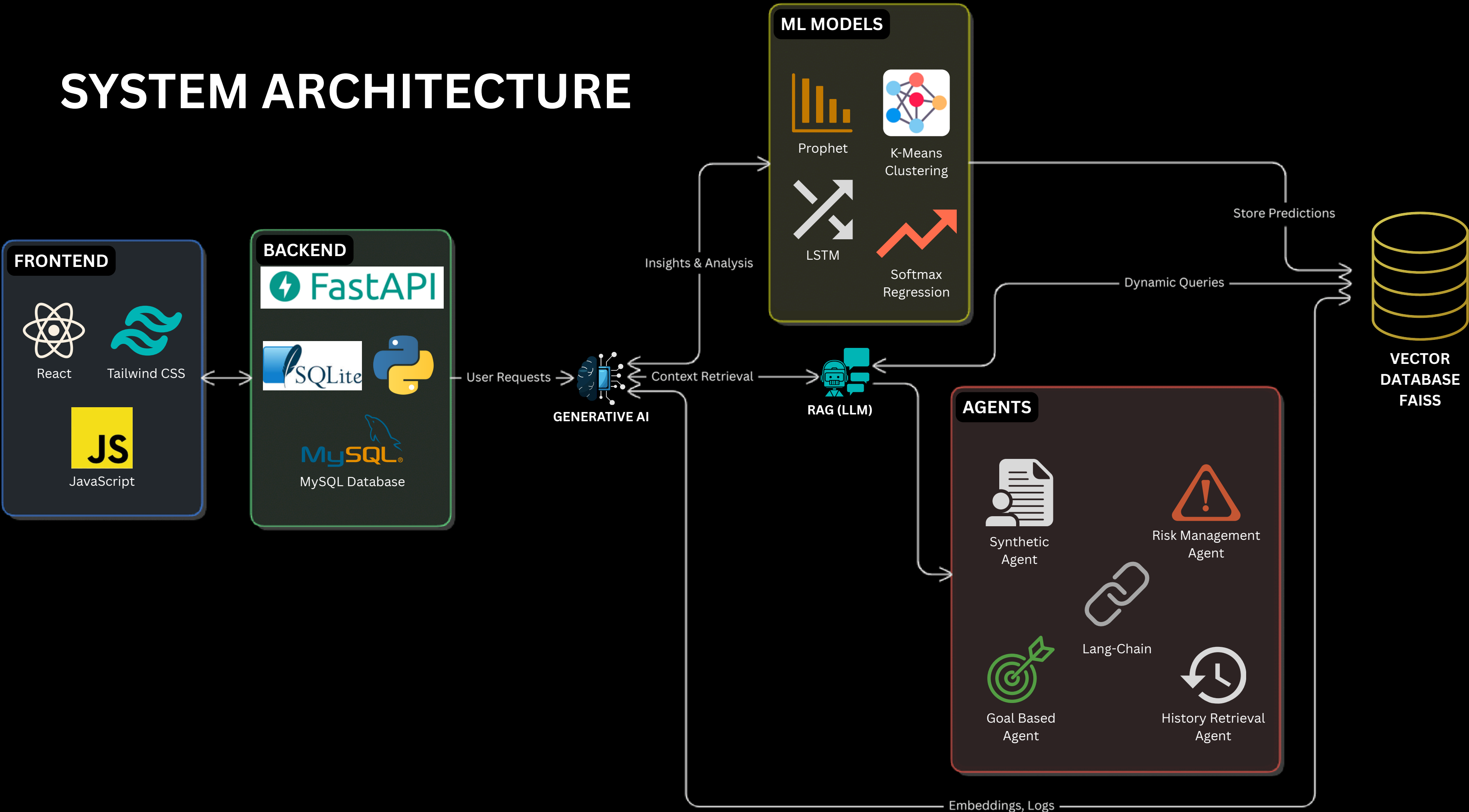
Role: Aggregates retrieved data, applies reasoning, and synthesizes a final response using the LLM.

4. History Agent

Purpose: Tracks and utilizes past interactions or user data.

Role: Retrieves historical data, maintains context across sessions, and personalizes responses based on user history.

SYSTEM ARCHITECTURE



VECTOR DATABASE

What is a Vector Database?

A Vector Database (VDB) stores embeddings (numerical representations of text/documents).

It enables semantic search by finding similarity between query vectors and stored document vectors.

Ideal for finance where large volumes of KYC forms, loan PDFs, policies, and market data need to be searched efficiently.

Role of FAISS:

- FAISS (Facebook AI Similarity Search) is the engine used for indexing and retrieval.
- Converts documents into embeddings and builds an index.
- Supports billions of vectors while keeping retrieval fast and memory-efficient.

Why FAISS is Fast & Efficient:

- Uses approximate nearest neighbor (ANN) search for lightning-fast retrieval.
- Optimized with CPU/GPU acceleration.
- Balances accuracy and speed, making it scalable for financial institutions.

Benefits in This Project:

Stores a large volume of financial docs (KYC, loans, policies, trends).

Provides real-time retrieval for RAG pipeline.

Efficient + scalable, making it optimal for banks/fintechs to deliver instant, context-aware financial plans.

FUTURE SCOPE

Open Banking APIs (like UPI, RBI Account Aggregator in India, PSD2 in EU, Plaid in US) to fetch user's financial data automatically.

Provide real-time financial health monitoring by syncing with credit cards, loans, and bank accounts.

Use Bayesian Portfolio Optimization to estimate probability distributions of returns and apply Monte Carlo simulations to test portfolio resilience under thousands of market scenarios.

CONCLUSION

FinPlan application leverages GenAI, agents, RAG, and LangChain to create an intelligent financial planning assistant that tailors advice based on a user's net worth, goals, and risk profile. By integrating clustering and softmax regression for portfolio allocation and using Prophet for gold price prediction, the system provides personalized and data-driven financial insights. This foundation demonstrates how AI and ML can work together to make financial planning more interactive, automated, and user-centric.

THANK YOU